

SURFACE SOIL SAMPLES — XRF READING DETAIL

Waste-vs-Native Surface Soil Investigation | Niton XL5 Plus, Soil Mode | 2 shots per sample

Date: 5-4-2026
Weather: Cloudy / Overcast / rain

Sampler(s): Mo. Sharif + Ariel Limberger
Project #: D3913200

Pg: 1/4
XRF Mode: Soil

IDENTIFICATION

FIELD READINGS

AVERAGES

Sample ID	Shot #	XRF Rdg # (Instrument)	Mode	Hg (ppm)	As (ppm)	Pb (ppm)	Time	Notes	AVG Hg (ppm)	AVG As (ppm)
XRF-OMB-1	1	1504	Soil	17	2	22	1304		15	1
	2	1505	Soil	13	0	23	1307			
XRF-OMB-2	1	1506	Soil	39	7	17	1310		41	7
	2	1507	Soil	43	7	19	1313			
XRF-OMB-3	1	1508	Soil	11	6	7	1317		10	5.5
	2	1509	Soil	9	5	11	1320			
XRF-OMB-4	1	1510	Soil	8	2	10	1322		7	3.5
	2	1511	Soil	6	5	9	1325			
XRF-OMB-5	1	1512	Soil	115	4	15	1329		114	5.5
	2	1513	Soil	113	7	13	1331			
XRF-OMB-6	1	1514	Soil	108	3	11	1335		105	3
	2	1515	Soil	102	3	12	1338			
XRF-OMB-7	1	1516	Soil	35	4	11	1345		36.5	4
	2	1518	Soil	38	4	10	1349			
XRF-OMB-8	1	1519	Soil	203	10	20	1353		200.5	10.5
	2	1520	Soil	198	11	20	1356			
XRF-OMB-9	1	1521	Soil	3851	10	51	1359		3781.5	10
	2	1523	Soil	3712	10	57	1405			
XRF-OMB-10	1	1524	Soil	952	20	20	1408		947	20.5
	2	1525	Soil	942	21	18	1412			
XRF-OMB-11	1	1526	Soil	134	10	6	1416		134	9
	2	1527	Soil	134	8	9	1419			
XRF-OMB-12	1	1528	Soil	6	0	5	1422		6	0
	2	1529	Soil	6	0	4	1426			
XRF-OMB-13	1	1530	Soil	32	70	12	1428		29.5	70.5
	2	1532	Soil	27	71	14	1433			
XRF-OMB-14	1	1534	Soil	7	0	18	1439		6.5	0
	2	1535	Soil	6	0	3	1442			
XRF-OMB-15	1	1536	Soil	112	54	9	1445		111	54
	2	1537	Soil	110	54	9	1447			

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Waste-vs-Native Surface Soil Investigation | Niton XL5 Plus, Soil Mode | 2 shots per sample

Date:

5-4-2026

Sampler(s):

Mo Sharif

April Linberger

Pg:

 $\frac{2}{4}$ **Weather:**

Cloudy / rainy / overcast

Project #:

D3913200

XRF S/N:

XRF Mode:

Soil

IDENTIFICATION

FIELD READINGS

AVERAGES

Sample ID	FIELD READINGS							AVERAGES		
	Shot #	XRF Rdg # (Instrument)	Mode	Hg (ppm)	As (ppm)	Pb (ppm)	Time	Notes	AVG Hg (ppm)	AVG As (ppm)
XRF - 0mb-16	1	1538	Soil	180	52	4	1450		180	52
	2	1539	Soil	180	52	4	1452			
XRF - 0mb-17	1	1542	Soil	212	40	53	1501		212.5	40
	2	1543	Soil	213	40	7	1505			
	1		Soil							
	2		Soil							
	1	1542	Soil				1501			
	2		Soil							
	1		Soil							
	2		Soil							
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	2		Soil							
	1		Soil							
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	2		Soil							

SURFACE SOIL SAMPLES — XRF READING DETAIL

Waste-vs-Native Surface Soil Investigation | Niton XL5 Plus, Soil Mode | 2 shots per sample

Date:

5/4/26

Sampler(s):

MO. SHARIF, ARIEL LINEBERGER

Pg:

3/4

Weather:

Cloudy + overcast

Project #:

D3913200

XRF S/N:

XRF Mode:

Soil

IDENTIFICATION

FIELD READINGS

AVERAGES

Sample ID	Shot #	XRF Rdg # (Instrument)	Mode	Hg (ppm)	As (ppm)	Pb (ppm)	Time	Notes	AVG Hg (ppm)	AVG As (ppm)
XRF-SWRC-1	1	1486	Soil	108	3	24	1158		109.5	3
	2	1487	Soil	111	3	25	1201			
XRF-SWRC-2	1	1489	Soil	82	0	16	1222		81.5	1
	2	1490	Soil	81	2	19	1224			
XRF-SWRC-3	1	1491	Soil	57	18	7	1226		59.5	19
	2	1492	Soil	62	20	8	1229			
XRF-SWRC-4	1	1493	Soil	45	0	7	1233		46.5	0
	2	1496	Soil	48	0	10	1242			
XRF-SWRC-5	1	1494	Soil	79	37	23	1237		79.5	41.5
	2	1497	Soil	80	46	21	1245			
XRF-SWRC-6	1	1498	Soil	826	0	18	1248		827.5	0
	2	1499	Soil	829	0	16	1251			
XRF-SWRC-7	1	1500	Soil	109	15	7	1253		110	15.5
	2	1501	Soil	111	16	7	1256			
XRF-SWRC-8	1	1502	Soil	52	2	6	1259		53.5	2
	2	1503	Soil	55	2	7	1301			
	1		Soil							
	2		Soil							
	1		Soil							
	2		Soil							
	1		Soil							
	2		Soil							
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	2		Soil							
	1		Soil							
	2		Soil							

Jacobs

Thermo Scientific

Date: 5/4/26

Weather: cloudy + overcast 65°F

Sampler(s): N. BROWN, M. SHARIF

Pg: 4

Project #: D3913200

XRF S/N: X501203

Run system health check each morning (col H). Shoot IARM 35NN puck once at start & end of day. Record measured values per element. Compare to certified values below. Hg is NOT in the alloy puck.

IARM 35NN CERTIFIED VALUES →

97.22%

10,860

4,780

3,310

1,740

1,250

 $\pm 20\%$ RPD

DAILY LOGISTICS

CHECK

ALLOY PUCK — MEASURED VALUES (one reading per event)

RESULT

ALLOY PUCK — MEASURED VALUES (one reading per event)															
Date	Time	Start/ End	Operator	Instrument ID / S/N	Mode	XRF Rdg #	Health Check (P/F)	Fe (%)	Cr (ppm)	Mo (ppm)	Mn (ppm)	Cu (ppm)	Ni (ppm)	Pass/ Fail	Notes
5/4/26	050	Start	MO	X501203	mining	1484	P	86.03	8722	4186	2720	1614	1509	P	
5/4/26	1645	END	MG	X501203	mining	1544	P	83.62	2220	3957	2802	1634	1185	P	Cr low - re-try ↓
5/4/26	1651	END	MO	X501203	mining	1545		85.57	8699	4093	2728	1655	1349	P	

FIELD NOTES:

VERIFICATION: If puck values are within $\pm 20\%$ RPD of IARM 35NN certified values and consistent between start/end, instrument is verified. If outside range $\rightarrow$ re-run health check, document corrective action.